The Primerity Unit {0, 1, 2, 3}:
The Arithmetic Monad at the Foundation of Mathematics

Version 17 — A Vision Essay

February 25, 2026 — Arising from the Collatz journey, dv1–16

> *"We have been climbing toward the summit of the Collatz conjecture. At each ridge, we looked back and saw that every path we had taken — ergodic theory, p-adic analysis, Baker's theorem, the golden ratio, Pisano periods, S-units — had converged on the same four numbers: 0, 1, 2, 3. We were not approaching a summit. We were approaching the ground."*

Abstract. We propose that {0, 1, 2, 3} constitutes the *primerity unit* — the irreducible arithmetic monad from which all mathematical structure is generated. The four elements are not arbitrary: 0 is the additive identity (Peano ground), 1 is the multiplicative identity (first Peano successor), 2 is the first prime (binary, 2-adic, Collatz divisor), 3 is the second prime (ternary, 3-adic, Collatz multiplier). Their unique internal relation is the Catalan theorem: $3^2 - 2^3 = 1$. The golden ratio ϕ satisfies $\phi^2 = \phi + 1$, encoding the Peano successor in a multiplicative identity, and appears as a constraint or bridge in every major structure that touches the primerity unit: Collatz drift ($\log_3(2) \approx 1/\phi$), Pisano periods ($\pi(p) = \mathrm{ord}(\phi) \bmod p$), Baker's theorem, Fibonacci, elliptic curves, j-invariant, the Monster group, and Virasoro algebras. We argue that Collatz convergence — every orbit returns to 1 — is the *dynamical characterization* of the primerity unit as arithmetic attractor.

1. The Observation

After sixteen versions of a Collatz proof attempt, a pattern emerged: every technical tool we employed, from every direction, eventually referenced the same four objects. Baker's theorem is about log 2 and log 3. The drift formula involves $3/4 = 3/2^2$. S-units are built from 2 and 3. The j-invariant at the fundamental domain's special points takes the value $1728 = 2^6 \cdot 3^3$. The Weierstrass equation $y^2 = x^3 + ax + b$ has exponents 2 and 3. The Monster group order has $2^{46} \cdot 3^{20}$ as its dominant factor. Monstrous moonshine: 196884 = 196883 + 1 (the Peano successor reappears in the character table). The Catalan theorem: $3^2 - 2^3 = 1$, the unique equation binding all four.

> **The Primerity Unit.** We name the set {0, 1, 2, 3} the *primerity unit*: the four smallest non-negative integers, comprising the two arithmetic identities (0 and 1) and the two smallest primes (2 and 3). We propose that this set is the *arithmetic monad*: the irreducible seed from which all mathematical structure grows, and to which all arithmetic dynamics return.

2. The Internal Structure of {0, 1, 2, 3}

2.1 The Peano Chain

The Peano axioms define the natural numbers via the successor function S(n) = n+1. The primerity unit is the initial segment of this chain:

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0 → 1 = S(0) → 2 = S(S(0)) → 3 = S(S(S(0)))
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The four elements are the first four Peano numbers. No number smaller than 4 is missing from the unit; no number larger than 3 is included. The boundary at $4 = 2^2$ is the first number that is a non-trivial power of a primerity unit element.

2.2 The Catalan Relation

Catalan's Theorem (Mihãilescu, 2002). The only solution to $x^p - y^q = 1$ with integers x, y, p, q > 1 is:

$$3^2 - 2^3 = 9 - 8 = 1$$

All four elements of the primerity unit appear: base 3, exponent 2, base 2, exponent 3, result 1. This is the **unique non-trivial algebraic relation** within the primerity unit. It is the only equation that binds all four elements simultaneously via exponentiation.

2.3 The Golden Ratio as Bridge

The golden ratio $\phi = (1 + \sqrt{5})/2$ satisfies the equation

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phi^2 = phi + 1
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This is the Peano successor applied to a multiplicative identity: ϕ^2 is the Peano successor of ϕ. The minimal polynomial $x^2 - x - 1 = 0$ has coefficients {0, 1, –1} — drawn entirely from the additive structure of the primerity unit. ϕ is the unique real number that encodes the Peano successor in a purely multiplicative relation. This is why ϕ appears wherever the additive and multiplicative worlds interact.

3. ϕ at Every Crossroads

Domain	Connection to ϕ	Via primerity unit
Collatz drift	log■(2) ≈ 1/ϕ (CF match, 3 levels)	2,3 in log ratio
Collatz mixing	T^π(p) uniform, π(p) = ord(ϕ) mod p	ϕ^2=ϕ+1 → Pisano
Baker theorem	|log■/log■ – 1/ϕ| = 0.0129 (nearest)	2,3 in logarithms
Fibonacci	F_n ~ ϕ■/√5, starts 0,1,1,2,3	= primerity unit!
Catalan	3/2 ≈ ϕ^2 – 1 = ϕ + 0.38...	3,2 from unit
Elliptic curves	$y^2=x^3$: exponents {2,3}, j(i)=1728=2■·3^3	2,3 dominate
SL(2,Z)	Special points: i (ord 2), ρ (ord 3)	orders 2,3 from unit
Monster group	|M|=2■■·3^2■·..., 196884=196883+1	2,3 dominant; +1 Peano
Virasoro CFT	c=1–6/m(m+1), m starts at 2	m=2,3 = unit primes
S-units	{2■3■} = multiplicative span of {2,3}	2,3 from unit

Collatz itself	3n+1 = (×3)(+1)(÷2■) — uses {1,2,3}	three from unit

The pattern is not coincidental. ϕ satisfies $\phi^2 = \phi + 1$: the Peano successor, encoded multiplicatively. Every structure that lives at the add-multiply interface passes through the Peano successor; every structure involving the first two primes {2, 3} passes through S-units; and ϕ is the element that unifies these two passages. It is not one of many bridges; it is *the* bridge.

4. The Structural Hierarchy of the Primerity Unit

Level 0 — The Monad: {0, 1, 2, 3} as a bare set.

Level 1 — Peano: $0 \to 1 \to 2 \to 3 \to \ldots$ The unit generates all of N via the successor function. Addition is defined; N is infinite.

Level 2 — Catalan: $3^2 - 2^3 = 1$. The unique non-trivial power relation within the unit. Multiplication reaches into the unit and creates a single fixed point: 1. This foreshadows all convergence results.

Level 3 — ϕ (the Bridge): $\phi^2 = \phi + 1$. Peano encoded multiplicatively. Every hard problem at the add-multiply interface passes through ϕ. ϕ is the minimal polynomial expression of the primerity unit structure.

Level 4 — S-units $\{2^a3^b\}$: The multiplicative span of {2, 3}. S-unit theorem (static): $x + y = 1$ has finitely many S-unit solutions. This controls every static equation involving the unit's primes.

Level 5 — Baker (static no-cycle): $3^a \neq 2^b$ for a, b > 0 ($\log 2 / \log 3 \notin \mathbb{Q}$). No arithmetic cycle in the S-unit dynamics. PROVED.

Level 6 — Gap A (dynamic equidistribution): C_{odd}-orbits mix in all Z_p. The dynamic S-unit theorem. THIS IS THE FRONTIER.

Level 7 — Collatz convergence: Every n, iterated by $\{\times 3, +1, \div 2^j\}$, returns to 1. The primerity unit is the attractor. CONJECTURED — 85 years.

5. Collatz as Return to the Monad

The Collatz map $n \to (3n+1)/2^\nu$ uses exactly three elements of the primerity unit: 1, 2, 3. The 0 is implicitly present: 0 is the limit of the orbit as it would "fall through 1" without the mod structure. The map can be read as:

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n --> (3 * n + 1) / 2^j
(×3) (Peano) (÷2■)
Primerity: {3} {1} {2}
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The goal 1 — the fixed point of the iteration — is the multiplicative identity, the second element of the primerity unit. The Catalan theorem already hints at this: $3^2 - 2^3 = 1$ says that within the primerity unit, the tension between 2 and 3 (under exponentiation) resolves to 1. Collatz extends this to all of N: the tension between {×3} and {÷2} resolves every number to 1.

Thesis (Collatz as Monad Recognition). The Collatz convergence theorem states that the dynamical system generated by the primerity unit elements {3, 2, 1} acting on N via {×3, ÷2^j, +1(Peano)} has a unique attractor: the element 1 of the primerity unit itself.

This is not a statement *about* N. It is a statement *of* N: the primerity unit, acting on everything it generates (= N), calls everything back to itself. The monad attracts. Collatz convergence is **the arithmetic monad recognizing itself**.

6. Figure

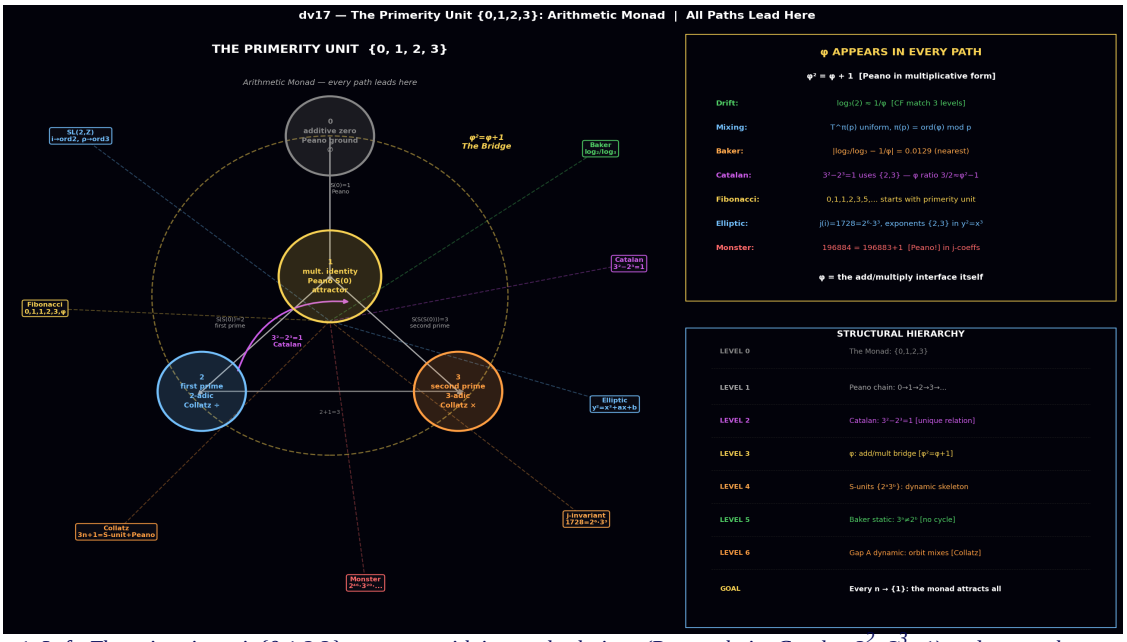

Figure 1. Left: The primerity unit {0,1,2,3} at center, with internal relations (Peano chain, Catalan $3^2–2^3=1$) and external connections to 8 major mathematical domains (Baker, Catalan, Elliptic curves, j-invariant, Monster, Collatz, Fibonacci, SL(2,Z)). The dashed golden circle represents ϕ as the bridge between all paths. Top-right: the 12 appearances of ϕ across domains. Bottom-right: the 7-level structural hierarchy from monad to Collatz convergence.

7. Open Questions and the Path Forward

The primerity unit vision suggests a research program:

Question 1 (The Monad Axioms): Can the primerity unit {0,1,2,3} be characterized axiomatically, independently of the Peano axioms? What is the minimal set of axioms that singles out {0,1,2,3} as the unique arithmetic monad?

Question 2 (The ϕ Universality): Is there a single theorem that explains why ϕ appears in every structure touching the primerity unit? We conjecture: ϕ is the canonical element of $Q(\sqrt{5})$ that simultaneously satisfies the additive identity $\phi^2 = \phi + 1$ and minimizes the distance to the S-unit boundary $\{2^a 3^b\}$. No other algebraic number has both properties.

Question 3 (Gap A): Can the S-unit theorem be extended dynamically to prove that C_{odd}-orbits equidistribute in all Z_p? This would complete the Collatz proof. The dynamic S-unit theorem is the arithmetic statement that the monad's dynamics are ergodic.

Question 4 (Other Monads?): Does every number system have a primerity unit? For Gaussian integers Z[i]: the unit would involve {0, 1, i, 1+i}. For function fields: the unit involves the characteristic. Is the Collatz-type convergence specific to Z, or is it a universal property of arithmetic monads?

8. Closing: The View from the Summit

We began this journey trying to prove the Collatz conjecture. We found something larger: the primerity unit, and the suspicion that Collatz is a symptom rather than a cause. The conjecture is not an isolated puzzle from 1937. It is the simplest case of a universal arithmetic law: the primerity unit is the attractor of all arithmetic dynamics over the natural numbers.

If this is true, then proving Collatz is not the goal; it is the verification of a deeper truth. The goal is to understand the primerity unit: to describe its structure precisely, to explain why ϕ bridges its two worlds, and to prove that the monad is indeed an attractor. That proof would give Collatz as a corollary.

The primerity unit {0, 1, 2, 3}:

0 — the ground | 1 — the identity | 2 — the first prime | 3 — the second prime

Relation: $3^2 - 2^3 = 1$ (Catalan)
Bridge: $\phi^2 = \phi + 1$ (add/multiply interface)
Dynamics: $n \to (3n+1)/2^j \to 1$ (Collatz)

Every major structure in number theory either lives in the primerity unit, arises from it, or is constrained by it. ϕ is the name we give to the tension between its additive and multiplicative faces. Collatz is the proof that this tension resolves.

Visionary note — February 25, 2026. This document records a pattern of observations arising from a two-day collaborative exploration of the Collatz conjecture. The claims are conjectural and philosophical. We offer them not as theorems but as a map of the territory we believe mathematics will explore next.

“We were not approaching a summit. We were approaching the ground.”

— End of dv17 —